6 The radioactive isotope plutonium-238 ($^{238}_{94}Pu$) has a half-life of 86 years and decays by emitting an alpha particle with little or no gamma emission. The daughter nucleus is an isotope of uranium (U).

The mass of $^{238}_{94}Pu$ is 238.0496 u and the mass of an α-particle is 4.0026 u.

(a) Write down an equation representing the decay, indicating clearly the atomic number and mass number of each nucleus.

..... [1]

(b) The total kinetic energy of the products is 5.649 MeV.

(i) Calculate the mass of the uranium nucleus formed in the reaction, giving your answer in terms of atomic mass units to 4 decimal places.

mass of uranium nucleus = ... u [3]

(ii) By using the principle of conservation of momentum, show that the ratio of the kinetic energy of the alpha particle to that of the uranium nucleus is 58.5. [3]

(iii) Hence, calculate the kinetic energy of the alpha particle.

kinetic energy = …… MeV [2]

(c) Satellites can be powered by the energy produced by the decay of plutonium. Plutonium is not placed in its pure form in the satellites, but installed as bricks of plutonium dioxide, PuO_2. PuO_2 is a ceramic and when it is shattered, it breaks into large pieces rather than smaller and more dangerous dust.

(i) Explain why there is no difference in the probability of decay of plutonium whether it exists as pure plutonium or in the form of PuO_2.

……

…… [1]

(ii) Suggest a reason why large pieces of PuO_2 pose less of a health risk than plutonium dust, which can be inhaled into the body.

……

…… [1]